SOEN 400: Data Mining & Analytics – Group Presentation Schedule

Since most groups are currently lagging behind in their projects, the next four weeks will be used to strengthen both your understanding of the Data Mining techniques and your progress on the group project. Each assigned group will prepare PowerPoint slides showcasing the Data Mining technique(s) and algorithms assigned to them, following the outline provided. The presentation should go beyond reading slides. You should explain and practically demonstrate the technique, using examples, calculations/formulas where applicable, applications, advantages and limitations, and most importantly, show how the technique will be applied to your group project.

Each group should select one member to represent the group during the presentation. The mark awarded to the presenter will be recorded as the mark for the entire group. Choose your representative carefully and ensure that they are fully prepared. The presenting group should also be prepared to respond to questions from both the class and the lecturer. Any member of the group may be called upon to answer, so every member should understand the technique being showcased.

By the end of the presentation, the group should be able to clearly demonstrate what the technique does, how it works, why it is suitable for their project, and how they intend to apply it. This activity will form part of your learning and group project assessment.

Wk	Groups	Topic	What you must cover
1	Groups 1, 5, 9 & 10	Classification Techniques	Supervised learning, classification concepts, training/testing data, decision trees, tree construction, entropy, information gain, pruning, KNN, distance measures, Naïve Bayes, Bayes theorem, conditional probability, comparison, applications, advantages and limitations
2	Groups 4 & 7	Regression & Predictive Analytics	Prediction vs classification, dependent/independent variables, regression concepts, simple & multiple linear regression, regression equation, intercept/slope, coefficient interpretation, prediction, assumptions, MAE, MSE, RMSE, R^2, applications and limitations

3	Groups 3 & 8	Clustering & Unsupervised Learning	Supervised vs unsupervised learning, clustering concepts/applications, similarity/distance measures, Euclidean & Manhattan distance, K-Means, centroid initialization, assignment, updating, convergence, hierarchical clustering, agglomerative/divisive approaches, dendrograms, evaluation and limitations
4	Group 6 & 2	Association Rule Mining	Transaction data, itemsets, frequent itemsets, association rules, Apriori, candidate generation, Apriori principle, minimum support, support, confidence, lift, interpretation, market-basket analysis, recommendations, strengths and limitations